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ASSIGNMENT



Subject: Front-end technology

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1. What is Figma? Explain its key features and why it is widely used in modern UI/UX design.

→ Figma is a premier cloud-based design and prototyping tool for creating user interface (UI) and user Experiences (UX) for websites and apps.

→ Figma works online in a web browser and can also be used as desktop application.

→ Many designers can work on the same design at same time which makes teamwork easy

Key features

- Real time Collaboration: Multiple team members can work on the same file.
- Cloud - Based / Browser Based: No installation is required, designs are automatically saved and accessible from any computer, making platform independent.
- Vector Design Tool: Advance vector network enable precise and detailed designing of interface.
- Prototyping: Designer can create interactive, clickable and animated prototypes to simulate the final product's user experience.
- Design Systems & Libraries: Enables creating, maintaining and sharing design components to ensure brand consistency.

Uses in Modern UI/UX Design

- Unmatched collaboration: The ability to work together simultaneously, leave comments and share links makes it superior for remote and cross-functional teams.

- Speed and Efficiency: It accelerates the design-to-development workflow through its all-in-one nature and developer-friendly feature.
- Platform Independent: Being browser-based, it runs on window, Mac, Linux without compatibility issues
- Free for individuals: Offers a robust free tier, making it accessible for beginners, students and freelancers.

2 Differentiate between Designing and Prototyping in Figma. Explain with one real life example.

Designing (Design mode)

- Focuses on the static visual appearance (layout, color, typography)
- Creating the visual look and UI component
- Elements like static shapes, text, images, components
- Static design
- Output High-fidelity mockups (static screen)
- Design Tab (Colors, strokes, layout)

Prototyping (Prototype mode)

- Focus on the interactive functionality (userflow, behaviour, animation)
- Simulating the user experience
- Elements like Connection (blue arrows), triggers, actions
- Interactive design (clicks, hovers, drags, smart animate)
- Output clickable, realistic simulations.
- Prototype Tab (interactions, device type)

Real World UI Example

1 Designing Phase (Static Visuals)

- Action: You create the Product Details screen frame
- Task: You define the product image, set the font size for the price, create a rounded rectangle for the "Add to Cart" button (using Auto Layout), and set the colors to match the brand
- Outcome: A beautiful, static image of what the screen will look like

2 Prototyping Phase (Interactive Flow)

- Action: You switch from the "Design" tab to the "Prototype" tab
- Task: You click the "Add to Cart" button and drag a blue line (noodle) to a new frame showing a Cart Successfully updated" popup.
- Settings: You set the trigger to "On Tap" and the animation to "Smart Animate" to make the button pulse slightly when pressed
- Outcome: A clickable simulation where a user can tap the button and see the pop-up, validating the user experience before it built.

3 What are Components and Styles in Figma? How do they help in maintaining consistency and saving time in large projects?

Components:-

Components are reusable UI elements (buttons, nav bars) with master-instance linking.

Created from layers, these include a Main Components (source) and instances (copies) Changes to the main Components propagates to all instance

Styles

→ Styles are saved visual properties (color, typography, effects)

→ Predefined colors, text styles and effects (shadows) applied across elements to ensure consistent branding.

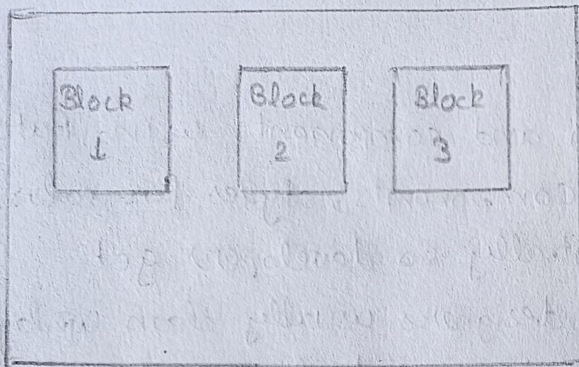
How They Save Time & Maintain Consistency

- Instant Updates : Changes a color style or a component's padding updates it everywhere instantly
- Scalability : Ideal for complex, multi-page projects as they eliminate the need to manually update repetitive elements
- Reduce ~~errors~~ : Using predefined styles ensures brand compliance and reduces design discrepancies
- Efficiency : Designers can drag and drop pre-built, standardized elements significantly reduce repetitive work.

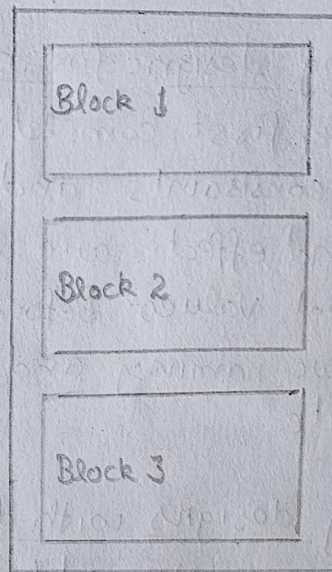
4 Explain Auto-Layout in Figma. How does it help in creating responsive designs for different screen sizes?

Auto-Layout in Figma is a property added to frames that allows elements within them to dynamically resize, reorder and reposition automatically based on content changes, padding and alignment settings. It enables responsive design by ensuring components (like buttons or cards) automatically adapt to different screen sizes, orientation and text length, eliminating the need for manual repositioning.

Layout 1



Layout -2



•→ How Auto Layout Helps Create Responsive Design.

- Dynamic spacing : Elements maintain consistent spacing, even if the container size changes
- Directional Flexibility : Layout can switch between vertical and Horizontal flows.
- Automatic Rearrangement : Removing or adding items automatically shifts the remaining elements into place
- Wrapping : Element can automatically wrap to a new line when the parent container becomes too small.

5 Describe the complete developer handoff process in Figma: How the designs shared, assets exported, and code snippets generated for developers.

The developer handoff process in Figma is a streamlined, real-time collaboration workflow that replaces traditional, manual spec handovers with a single source of truth - The Figma file itself. It centers on utilizing Figma's "Dev Mode" to inspect measure, and export assets directly from design components.

The end-to-end developer handoff process in figma

1. Finalizing designs for handoff

Designers first complete all screen and components using Auto Layout, constraints and reusable components, styles for color, text, and effects are defined centrally so developers get consistent values. Before handoff, designers usually clean up layers, use clear naming and ensure spacing and alignment are accurate.

2. Sharing designs with developers

Designs are shared by giving developers view or dev access to the figma file or via a shareable link. Developers can open the file in the browser or figma app without editing rights. Pages are often organized.

3. Inspecting designs (Dev mode)

Using Figma Dev Mode, developers can click any element to inspect

- Exact dimensions, spacing and positioning.
- Typography (font, size, weight, line height)
- Color (HEX, RGB, CSS variables)
- Auto Layout behavior

4. Exporting assets

Designers mark layers or components as exportable. Developers can then:

- Download assets (SVG, PNG, JPG, PDF)
- Choose multiple resolution
- Export icons and images directly from Figma.

5. Generating code snippets

Figma automatically generates ready-to-use code snippets for selected elements, including

- CSS (Flexbox, spacing, color, fonts)
- iOS (SwiftUI / UIKit)
- Android (XML / Compose)

6. Using Components and design tokens

Developers reference design token (color, typography, spacing) defined in Figma styles as variables. This helps map designs to CSS variable, Tailwind configs, or theme files keeping files design and code in sync.

7. Comments and collaboration

Developers can leave comments directly on designs to ask questions or report issues. Designers reply or update the UI in real time, creating a fast feedback loop and reducing misunderstanding.

8. Version history and updates

Figma's version history allows teams to track changes, roll back designs and see what was updated after handoff.

9. Final handoff checklist

- All screens finalized and approved
- Components and styles used consistently
- Assets marked for export
- Dev Mode access enabled
- Naming conventions followed